Module 2 Simulation: Hydroponic Systems Lab

What this simulation does

The Hydroponic Systems Lab puts you in the role of a mission engineer designing a hydroponic grow chamber for a space habitat. You configure three variables for a selected crop and run the system to see the outcome. The simulation has five scenarios, all available from the start. There is no single correct answer — different crop and configuration combinations produce different outcomes, and your job is to figure out which combinations work and why.

A spotlight tutorial runs on first load, walking you through the interface. You can dismiss it and return to it at any time.

The three variables you control

Lighting Spectrum: LOW, MED, or HIGH. Controls the intensity of the LED grow lights in the chamber. Different crops need different light intensities at different growth stages.

Nutrient Delivery: LOW, MED, or HIGH. Controls the concentration of nutrients delivered to the root system. Too little starves the plant. Too much causes nutrient burn.

Water Flow: LOW, MED, or HIGH. Controls how frequently water circulates through the system. Some crops tolerate dry cycles. Others need constant moisture.

How to run a scenario

Select a crop from the scenario list on the left panel. Each scenario presents a crop with a brief description of its growing requirements and the space mission context it serves.

Set your three variables using the LOW, MED, and HIGH buttons for each one. Read the crop description first — it will give you clues about what the plant needs.

Click RUN SYSTEM to see the outcome. The simulation will show you whether the crop thrived, struggled, or failed, and explain which variable caused the result.

Adjust your settings and run again. You are not penalized for failed runs. The simulation is designed to be explored, not solved on the first attempt.

What the outcomes tell you

A Thriving outcome means your configuration met the crop's needs. Read the explanation to understand which variable was most important.

A Struggling outcome means one or more variables were off. The feedback will identify which one and explain what the plant was missing or receiving in excess.

A Failed outcome means the configuration was significantly mismatched. The feedback explains the failure mechanism — what happened to the plant biology when conditions fell outside the tolerance range.

Connecting this to your assignment

Your System Design Brief asks you to describe the crop you chose, the configuration you settled on, the reasoning behind your choices, one limitation of your design, and one question your experience raised. Run each scenario at least twice: once with your initial guess, and once after you have read the feedback and adjusted. The second run gives you more to write about.

If you get stuck

Go back to the crop description. It usually tells you whether the plant is a heavy feeder, drought-tolerant, or light-demanding. Match those clues to the variable settings. If you cannot get a crop to thrive after three attempts, that failure is still useful: your brief can describe what you tried, what failed, and what you would test next if you had more time.